VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

Exploratory Data Analysis

AIML231 Week 4

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Today’s Lecture

What is EDA?

EDA methods by data type

Step-by-step EDA

Considerations in EDA

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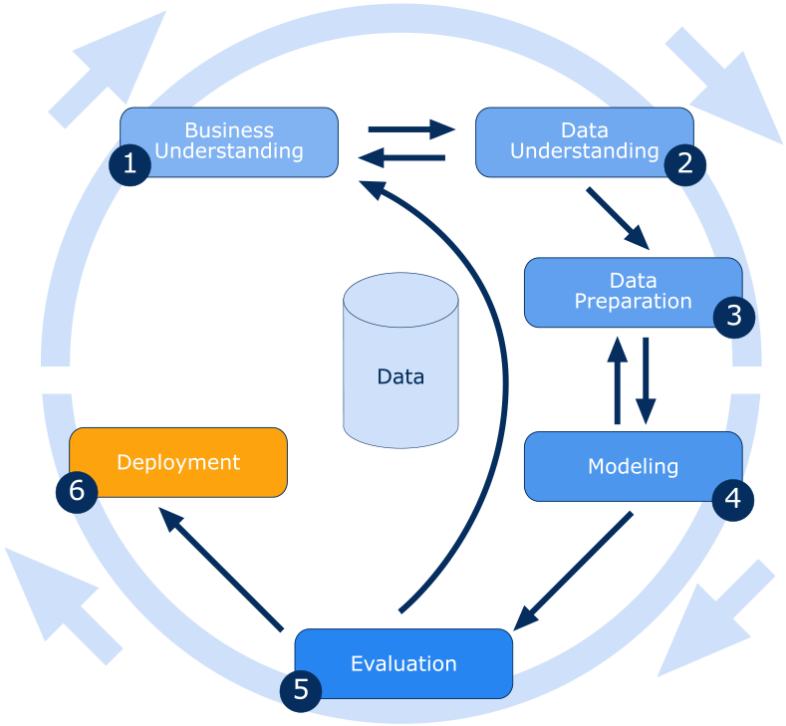
EDA methods by data type

Step-by-step EDA

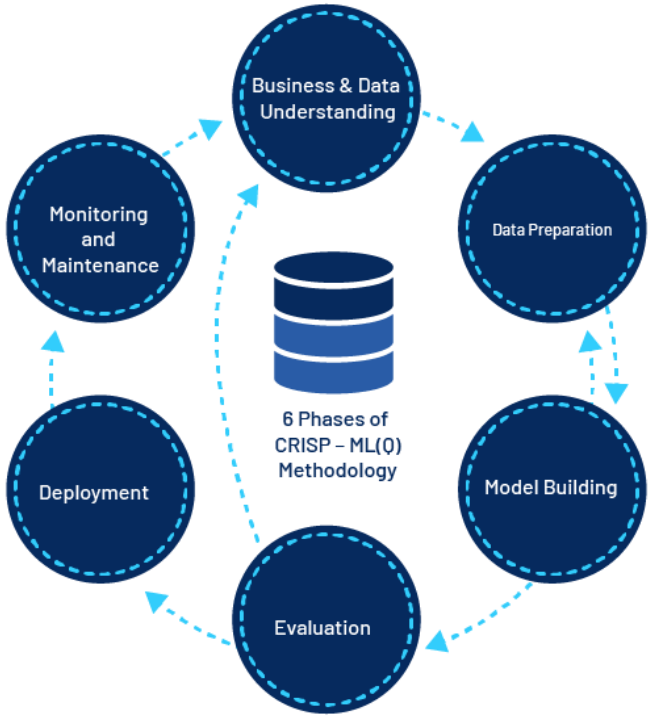
Considerations in EDA

Data Understanding and Preparation ☝️

- Exploratory Data Analysis (EDA) is a fundamental part of data understanding/data preparation
- Makes up ~80% of a project!

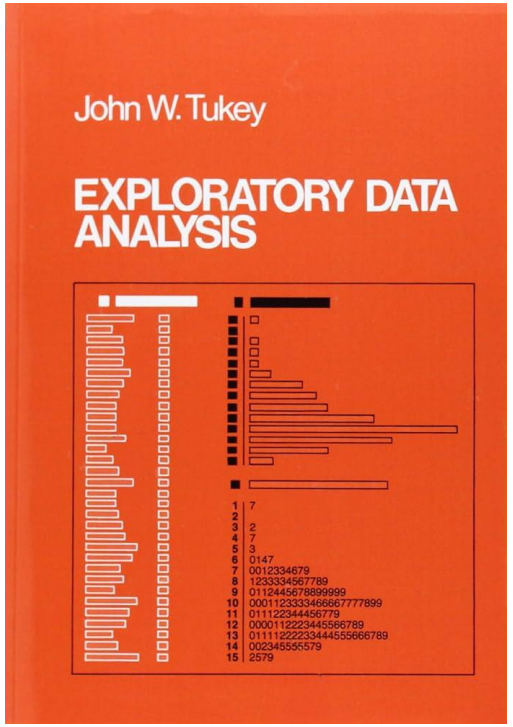


CRISP-DM



CRISP-ML(Q)

Exploratory Data Analysis (EDA)

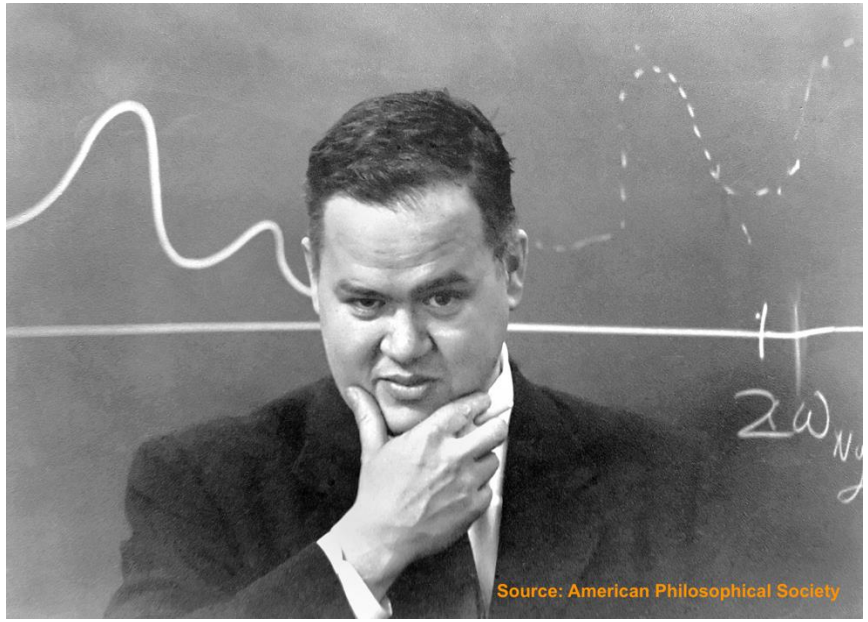


- EDA coined in 1977 by John Tukey
- Promoted a distinction between:
  - *Exploratory* analysis (what is my data?)
  - *Confirmatory* analysis (does my data confirm my hypothesis?)

If we need a short suggestion of what exploratory data analysis is, I would suggest that

1. It is an attitude, AND
2. A flexibility, AND
3. Some graph paper (or transparencies, or both).

- We Need Both Exploratory and Confirmatory, John W. Tukey



EDA Questions

EDA helps us understand a dataset before we model it. We do this by asking questions and…

Summarising

- How many variables are there?
- What types of variables (numeric, string, binary…)?
- How many samples are there?
- Computing statistics (mean, median, std, skew)

Visualising

- Plot the data – is it what you expected?
- Use histograms/box plots/line charts/heatmaps/…
- For 2d/3d data, plot spatially
- For higher dimensional data, dimensionality reduction

Familiarising

- What is the range of this data?
- Is it what you expected?
- Are some variables correlated with one another?
- Are there any redundant variables?
- What are the units of the data?

EDA Steps

- There isn't a set of concrete steps to follow for EDA as every dataset and task require different analysis!
- In general, EDA is an iterative cycle:
  - Generate questions about the data
  - Answer those questions by summarising and visualising the data
  - Use what you learn to refine your questions, or to generate new questions

- What type of variation occurs within my variables?
- What type of covariation occurs between my variables?
- Is my data formatted how I expected it to be?

Objectives of EDA

- Understand the structure and quality of the data
- Discover patterns and relationships
- Spot outliers and anomalies
- Generate preliminary hypotheses
- Check assumptions

EDA's main objective is *not* about making plots, but deciding what you should/can do next

What questions should EDA answer?

What does an “average” data point look like?	Mean, median, mode
How does the data vary from this average?	Variance, standard deviation
How is the data distributed?	Q-Q plots, skewness, kurtosis, plots
Are there outliers?	Box-plots, z-scores, interquartile range, scatter plots
Are the variables correlated?	Correlation scores, plots

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Types of EDA Methods

Graphical or non-graphical

Graphical: summarising data in a visual way, e.g., plots, histograms

Non-graphical: calculation of summary statistics

Univariate or multivariate

Univariate: analysing one feature

Multivariate: analysing several features (and often the relationships between them)

Qualitative or quantitative

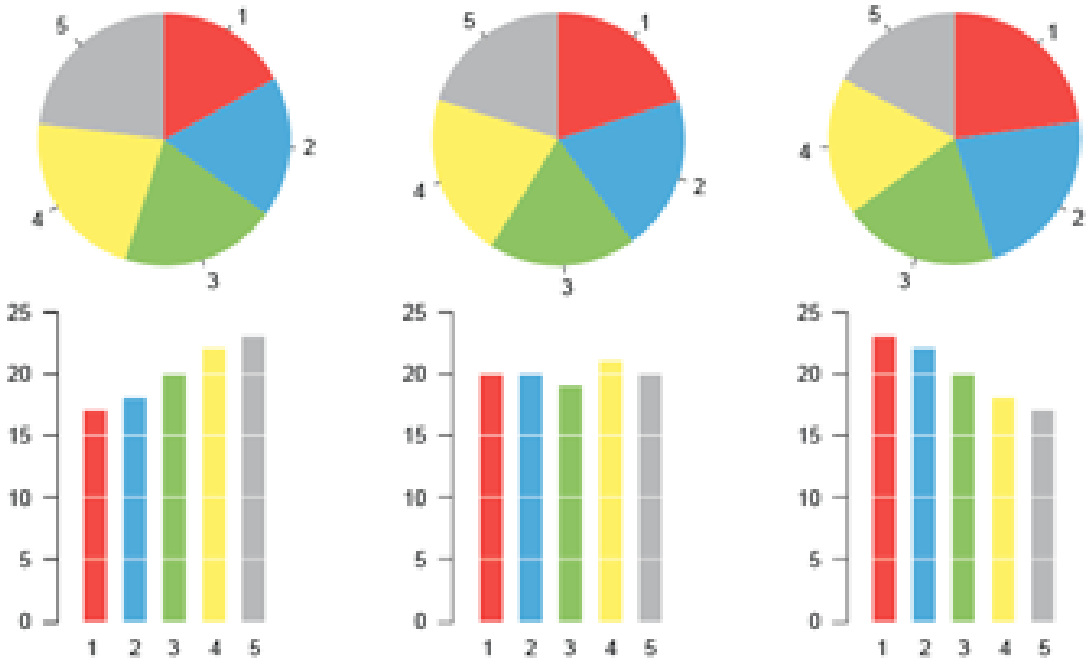
Qualitative: non-numerical data, e.g., hair colours, grades

Quantitative: numerical data

EDA Methods: Univariate, Graphical

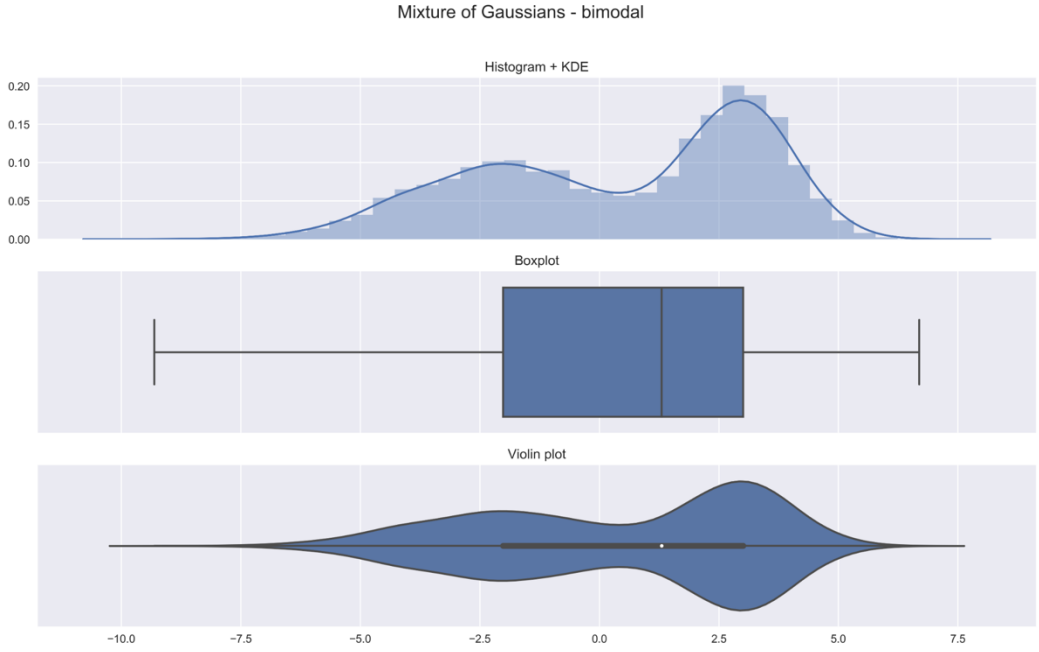
Qualitative variables:

- Bar charts
- Pie charts



Quantitative variables:

- Histogram
- Box/violin-plots



EDA Methods: Univariate, Non-Graphical

Qualitative variables:

- Frequency tables
- Mode

Hair colour	Frequency
Blonde	9
Red	4
Brown	13
Black	15
Blue	1
Grey	3
Total	**45**

Quantitative variables:

- Averages (mean, median, mode)
- Spread (variance, standard deviation)
- Outliers
- Distribution shape (skewness, kurtosis)

$$\text{(Mean) } \mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

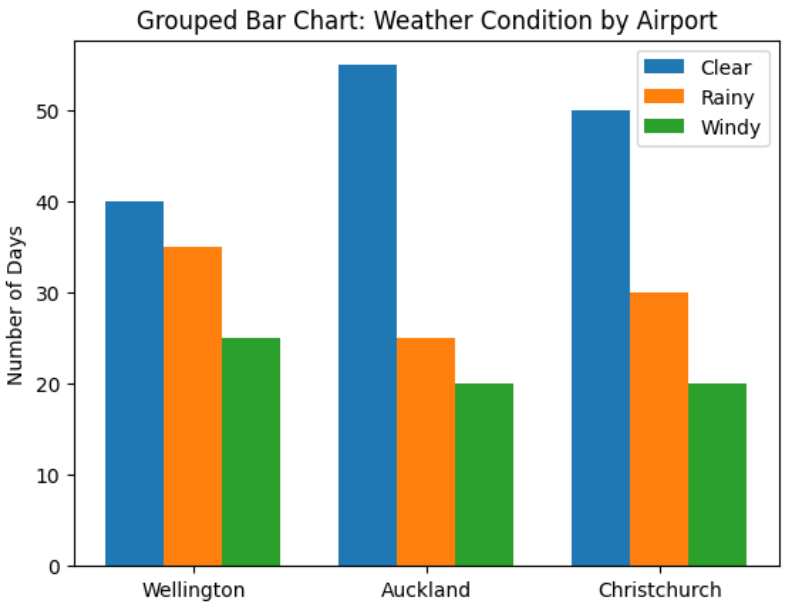
$$\text{(Variance) } \sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

$$\text{(Standard Deviation) } \sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

EDA Methods: Multivariate, Graphical

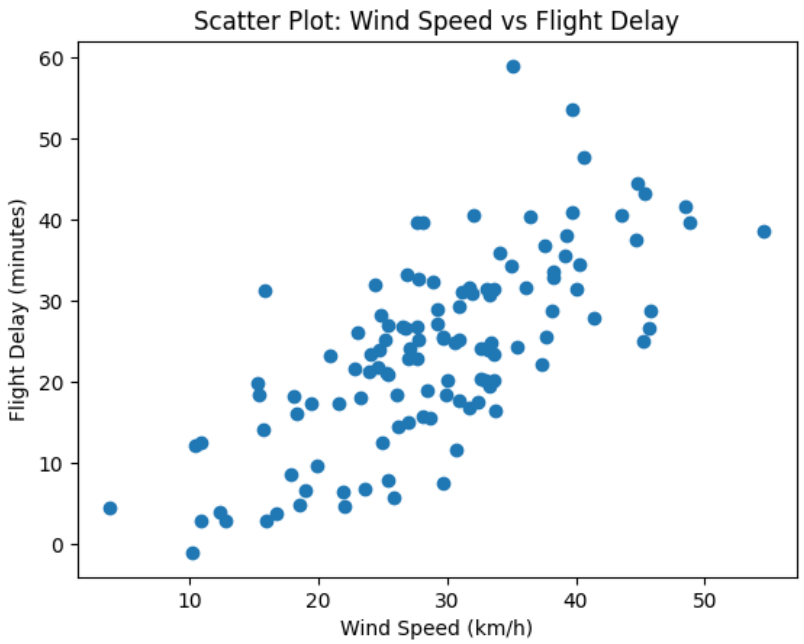
Qualitative variables:

- Grouped bar charts



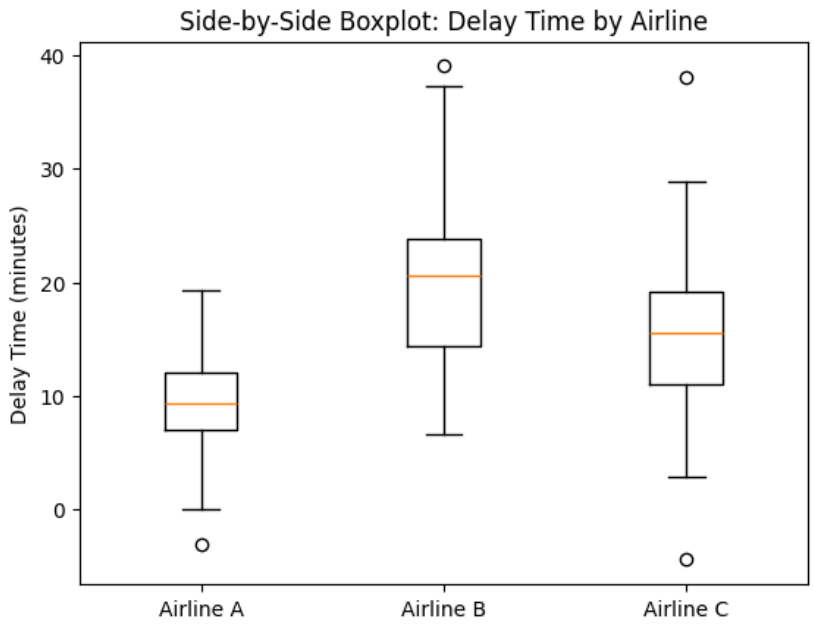
Quantitative variables:

- Scatterplots
- Correlation heatmaps



One qualitative, one quantitative variable:

- Side-by-side boxplot



EDA Methods: Multivariate, Non-graphical

Qualitative variables:

- Contingency tables

	On time	Delay-ed	Totals
WLG	80	20	100
AKL	60	10	70
CHC	45	30	75
Totals	185	60	245

Quantitative variables:

- Summary statistics
- Correlation/covariance

Descriptive Statistics

Variable	Obs	Mean	Std.Dev.	Min	Max
price	74	6165.257	2949.496	3291	15906
mpg	74	21.297	5.786	12	41
rep78	69	3.406	.99	1	5
headroom	74	2.993	.846	.846	5
trunk	74	13.757	4.277	5	23
weight	74	3019.459	777.194	1760	4840
length	74	187.932	22.266	142	233
turn	74	39.649	4.399	31	51
displacement	74	197.297	91.837	79	425
gear_ratio	74	3.015	.456	2.19	3.89
foreign	74	.297	.46	0	1

One qualitative, one quantitative variable:

- Statistics by group

	Mean Delay	Std Dev Delay
Airline A	9.47	2.25
Airline B	20.12	15.5
Airline C	11.3	7.3

Types of EDA Methods

	Univariate	Multivariate
Graphical	**Qualitative variable:** bar charts, pie charts (carefully!) **Quantitative variable**: histogram, box-plots, violin plots, stem-and-leaf plot	**One qualitative and one quantitative variable:** side-by-side boxplots, ... **Two or more qualitative variables:** grouped bar chart **Two or more qualitative variables:** scatterplot, correlation heatmaps, ...
Non-graphical	**Qualitative variable**: frequency table, mode **Quantitative variable**: calculate averages, spread, outliers, distributions, ...	**Qualitative vs quantitative variables:** group means/medians, difference in means, ... **Two qualitative variables:** contingency tables **Two (or more) quantitative variables:** correlation, covariance, summary statistics, ...

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What is EDA? ✅

EDA methods by data type ✅

Step-by-step EDA

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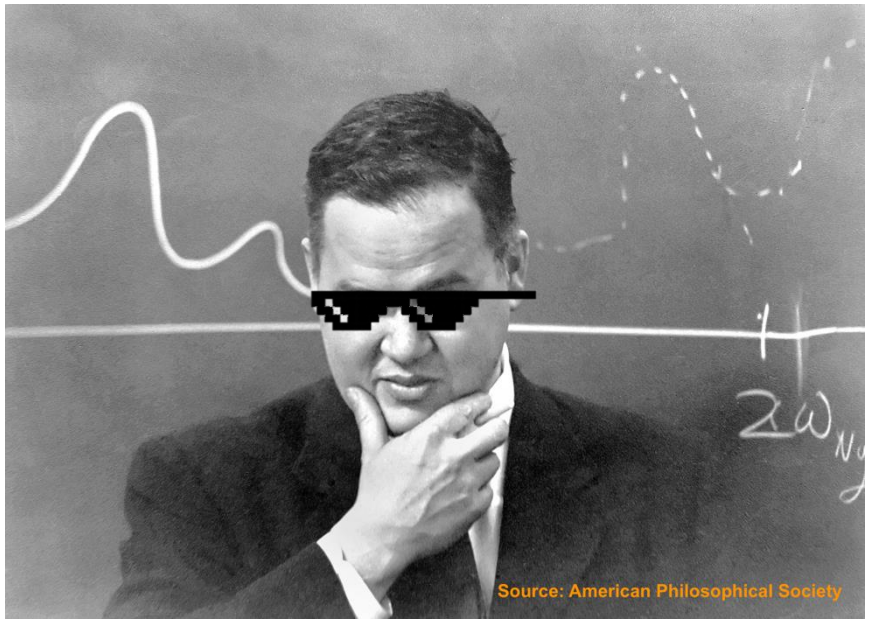
EDA Step by Step

A general overview of EDA… remembering that '*EDA is an attitude*'.

1. Inspect the data
2. Handle missing values
3. Explore data characteristics
4. Perform data transformations



Load and inspect the data

- Organise the data, often into a tabular format, made up of
  - Variables: usually the columns
  - Samples/instances/data points: usually the rows
- Look at the types of variables:
  - Qualitative attributes represent data in discrete categories
    - Ordinal data: an ordering to categories
    - Nominal data: no inherent ordering to categories
  - Quantitative attributes represent data on numerical, measurable scales

Major	Year of Study	Study Hours /Week	Exam Score %	Stress Level
ENGR	1	12	65	Medium
COMP	1	8	61	Low
COMP	3	15	N/A	High
AIML	2	20	81	Medium
ENGR	4	14	64	Low
MATH	N/A	17	77	Medium
AIML	1	11	61	High

Reasons for missing data

- Missing completely at random (MCAR): Missingness is unrelated to any data. *Example:* A completed survey is lost in the mail.
- Missing at random (MAR): Missingness depends only on observed variables. *Example:* Younger respondents are less likely to answer income questions - depends on age (observed), not on income.
- Missing not at random (MNAR): Missingness depends on the missing value itself. *Example:* Individuals with very low (or very high) income are less likely to report their income.

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Handle missing values

MCAR:

- Sample deletion: remove samples with missing data
- Variable deletion: remove variables with missing data

MAR/MNAR:

- Simple imputation: substitute in the mean, median, or mode of the variable
- Predictive imputation: use regression or classification models to predict and fill missing values

Depends on how many missing values there are!

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Univariate non-graphical

- Frequency tables

ENGR	COMP	AIML	MATH
2	2	2	1

Low	Medium	High
1	3	2

- Summary statistics for quantitative data

Variable	Mean	Std
Study hours	13.14	4.99
Exam score	68.17	7.88

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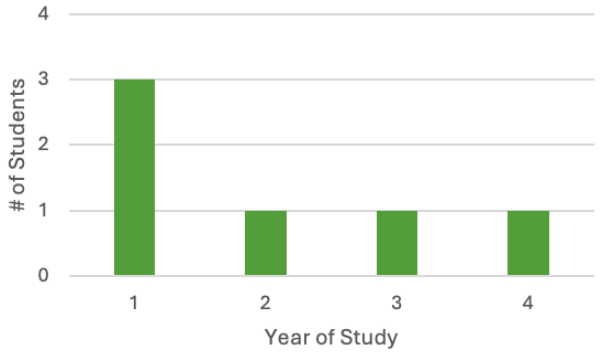
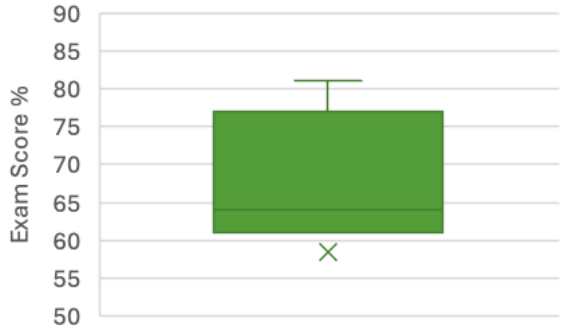
Univariate graphical

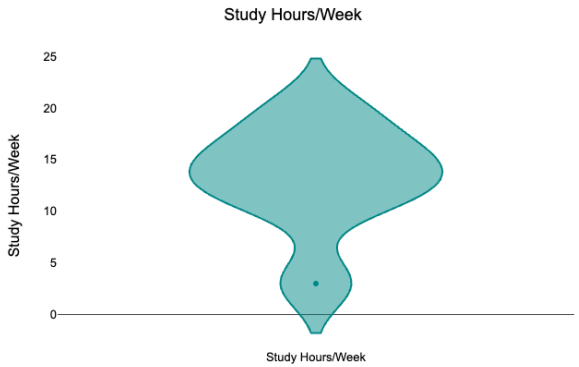
- Bar charts



- Box plots



- Violin plots



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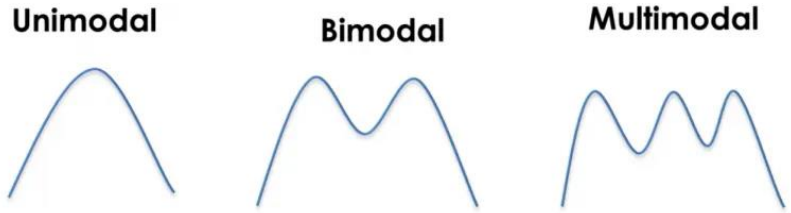
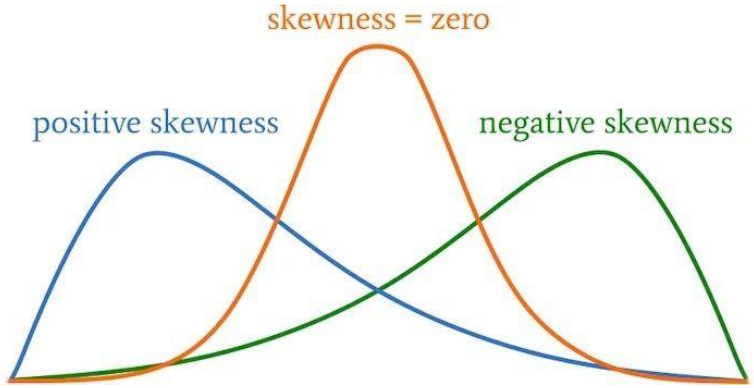
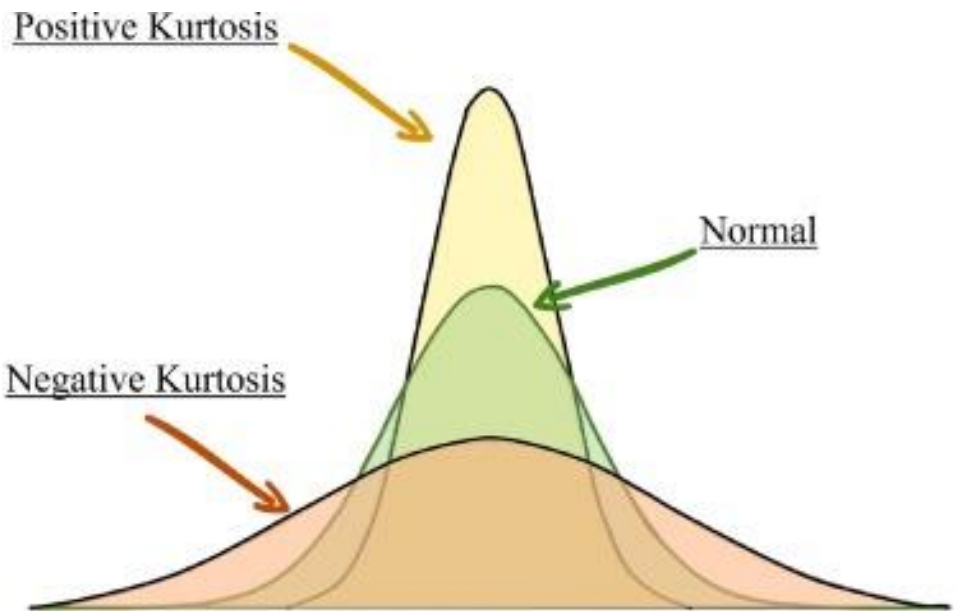
Univariate graphical



Plotting univariate data:

- Is data normally distributed?
- Is it unimodal/bimodal/multimodal?
- Skewness
  - Left (negative) skew: long tail on the left
  - Right (positive) skew: long tail on the right
- Kurtosis: tail heaviness
  - Positive kurtosis: heavier tails, more extreme values
  - Negative kurtosis: lighter tails, fewer extreme values



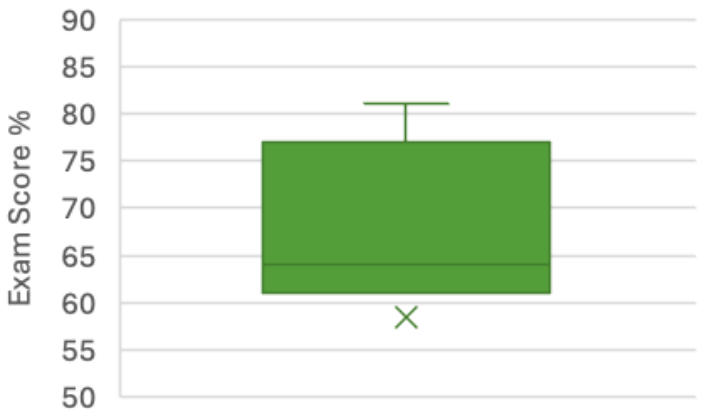


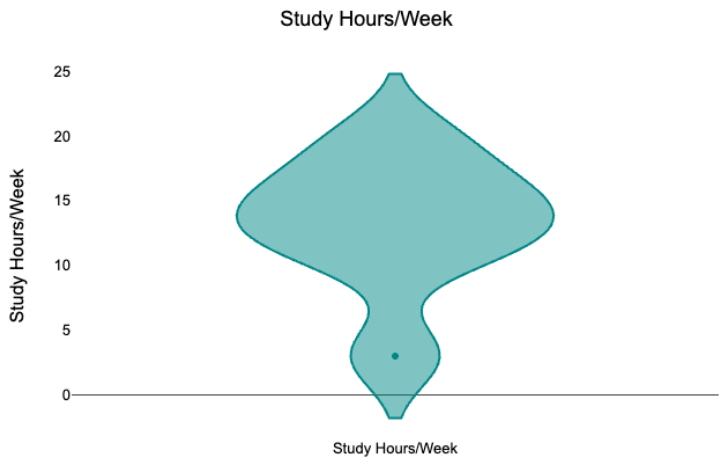
Univariate analyses

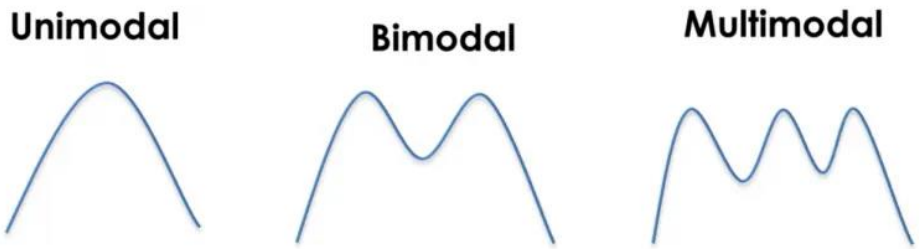
What am I looking for in these univariate analyses?

- Distribution of data variables
- Outliers
- Standard ranges of variables

Variable	Mean	Std
Study hours	13.14	4.99
Exam score	68.17	7.88







Multivariate non-graphical

- Contingency table

  (two qualitative variables)

Major	Low	Med	High	*Total*
ENGR	1	1	0	*2*
COMP	0	0	1	*1*
AIML	0	1	1	*2*
MATH	0	1	0	*1*
Total	*1*	*3*	*2*	*6*

- Grouped statistics

  (one quantitative, one qualitative)

Major	Mean SH	Std SH
ENGR	13	1.41
COMP	9	8.49
AIML	15.5	6.36
MATH	17	N/A

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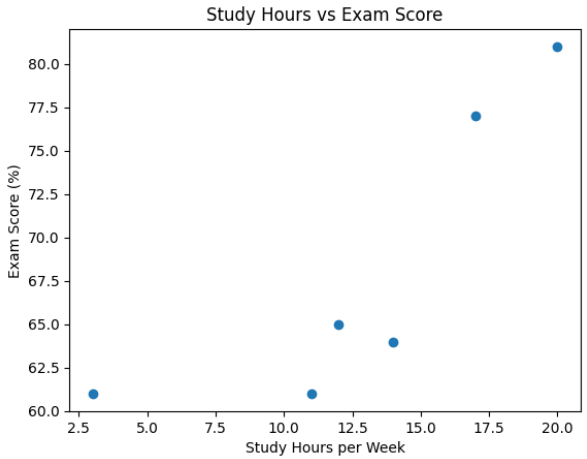
Multivariate graphical

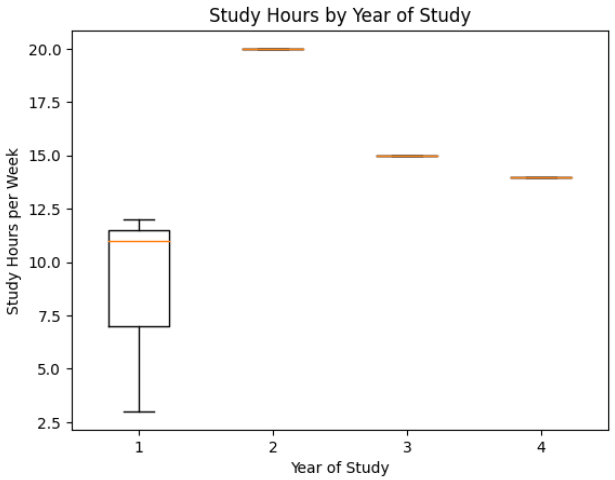
- Scatter plot

  (two quantitative variables)



- Side-by-side boxplot

  (one quantitative, one qualitative)



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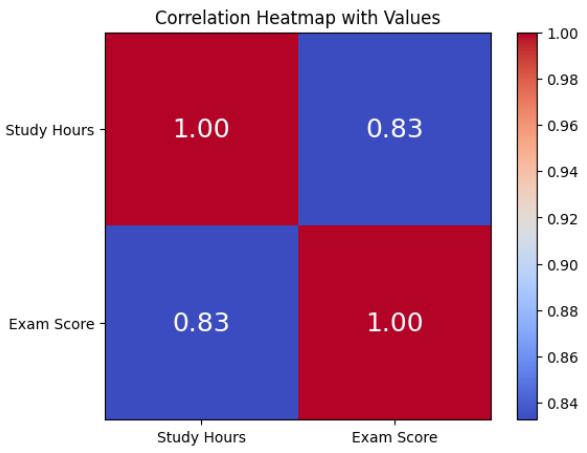
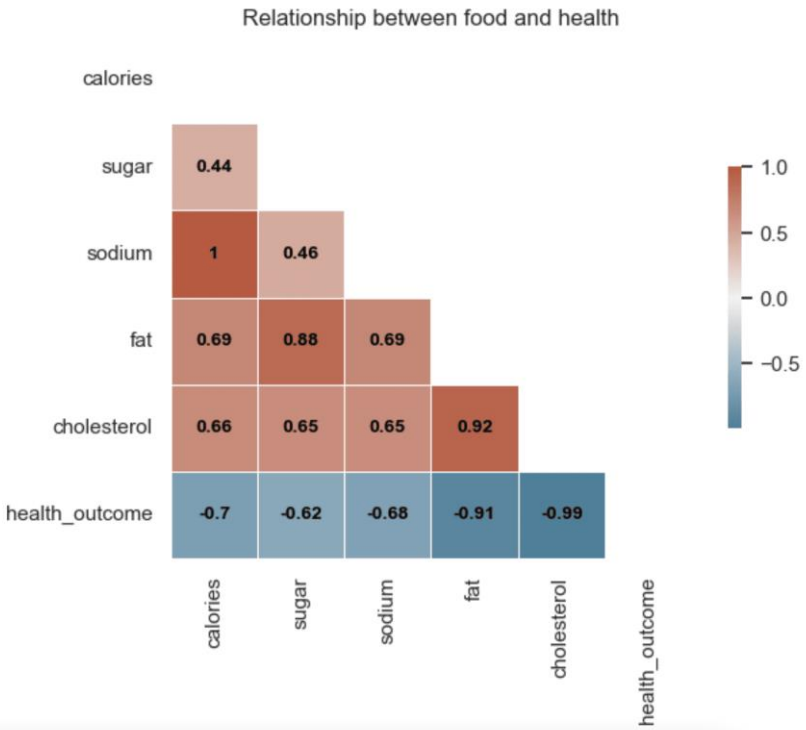
Multivariate graphical

- Correlation heatmap

  (multiple quantitative variables)





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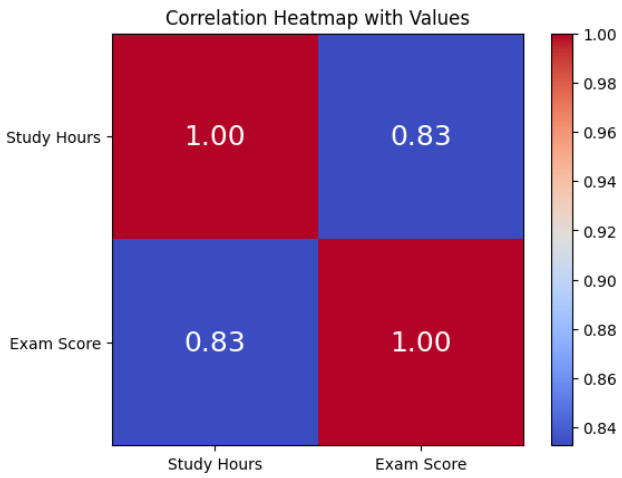
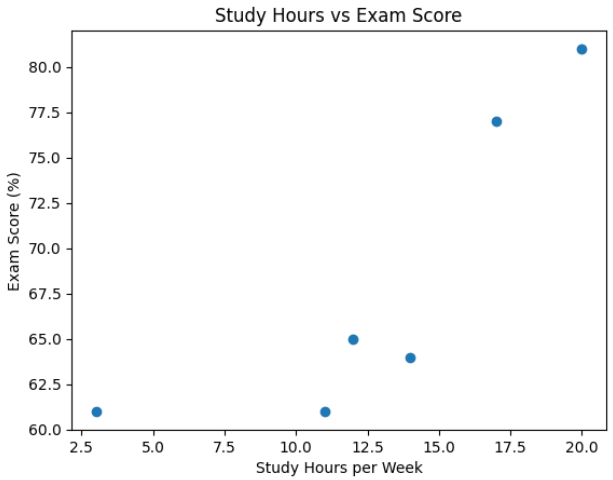
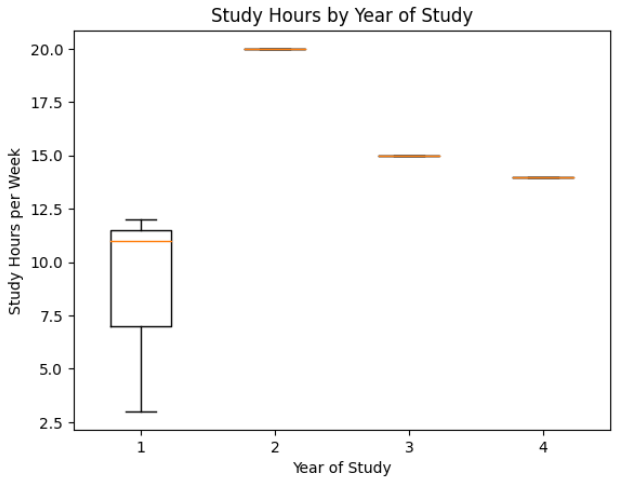
Multivariate graphical

What am I looking for in multivariate analyses?

- Relationships between variables
- Differences between groups







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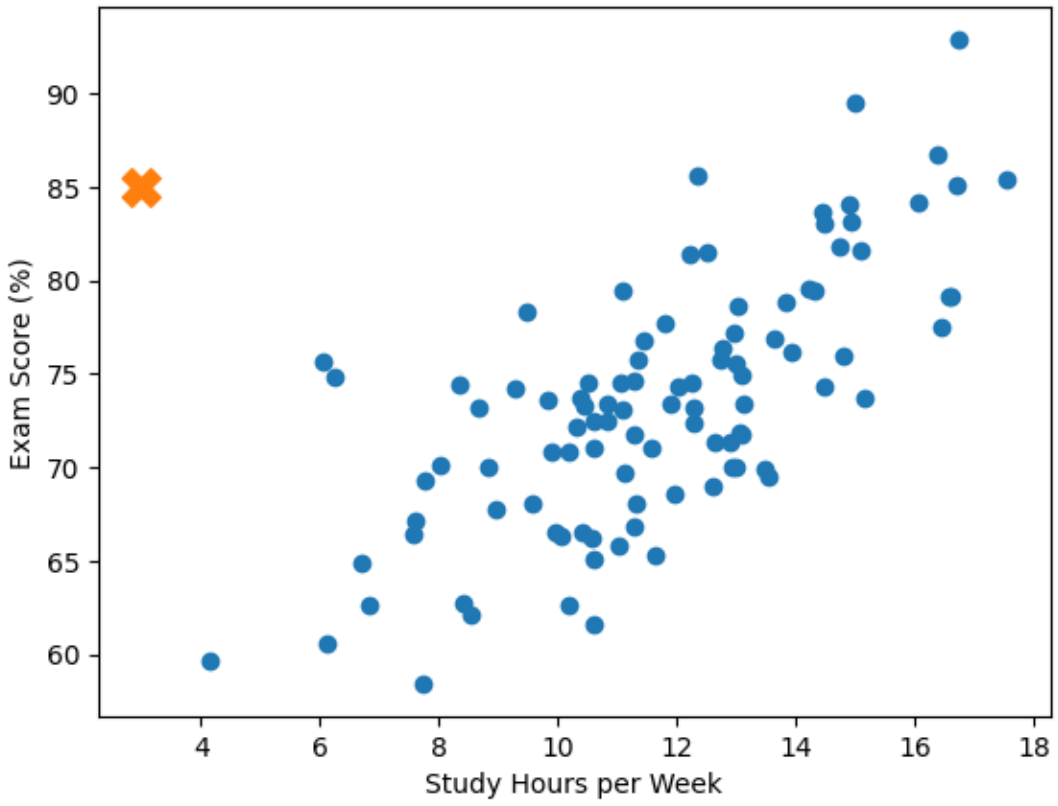
Finding outliers

Some observations may look normal when examining each variable separately, but become unusual when variables are considered together.

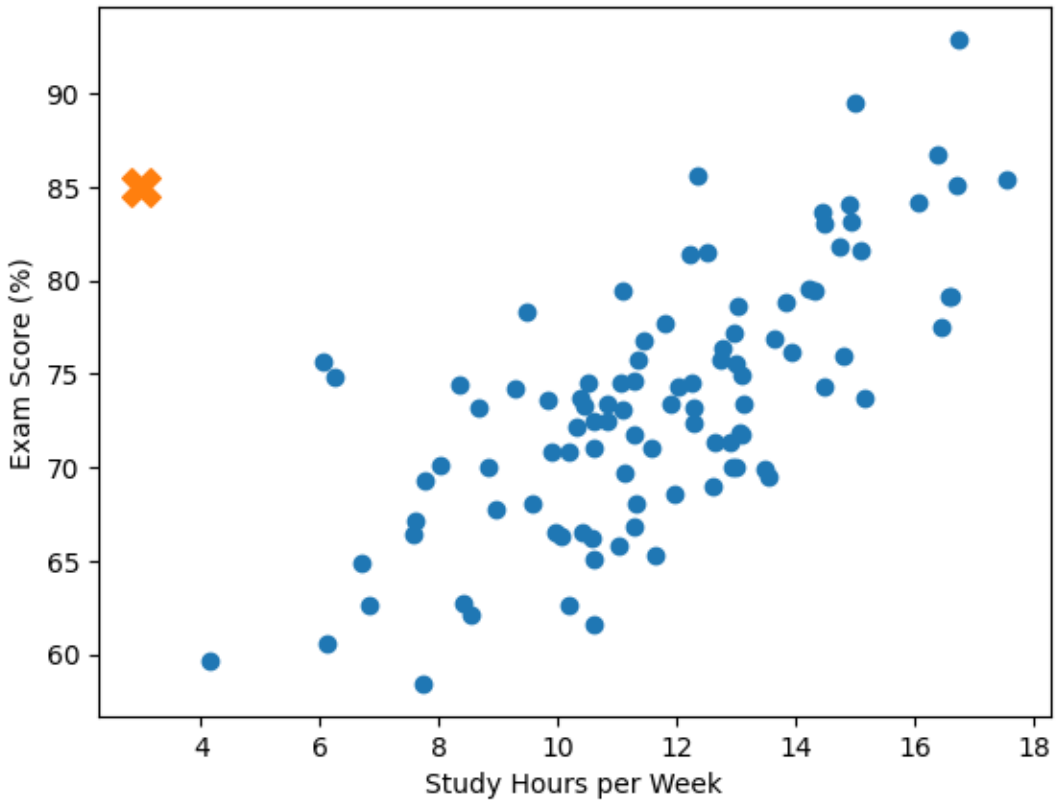
- A student who *studies 3 hours* per week is not unusual.
- A student who *scores 85%* is not unusual.
- But a student who studies 3 hours and scores 85% may be unusual relative to the overall pattern.



Dealing with outliers

Outliers are *not always mistakes*!

- Is it a typo? Is someone's height 1700cm rather than 170cm?
- A unit mismatch? 170ft rather than 170cm?
- A rare but valid observation? Perhaps that student really did get 85% with 3 hours of study per week…



If it is a confirmed error, remove the outlier by sample deletion or imputation.

Transformations for quantitative data

Mean-std scaling/z-score scaling/standardization:

- Produces mean 0, variance 1
- Sensitive to outliers

$$y = \frac{x - mean}{std}$$

Min-max scaling:

- Gives $y \in [0, 1]$
- *Very* sensitive to outliers:

$$y = \frac{x - min}{max - min}$$

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COMP	1	3	61	N/A
COMP	3	15	N/A	High
AIML	2	20	81	Medium
ENGR	4	14	64	Low
MATH	N/A	17	77	Medium
AIML	1	11	61	High

Transformations for qualitative data

One-hot encoding: Convert **nominal** variables into multiple binary variables, where each category has its own column

Major	Year of Study	Study Hours /Week	Exam Score %	Stress Level
ENGR	1	12	65	Medium
COMP	1	3	61	N/A
COMP	3	15	N/A	High
AIML	2	20	81	Medium
ENGR	4	14	64	Low

Major_ ENGR	Major_ COMP	Major _AIML	Major_ MATH	Year of Study	Study Hours /Week	Exam Score %	Stress Level
1	0	0	0	1	12	65	Medium
0	1	0	0	1	3	61	N/A
0	1	0	0	3	15	N/A	High
0	0	1	0	2	20	81	Medium
1	0	0	0	4	14	64	Low

Transformations for qualitative data

Ordinal encoding: Convert **ordinal** variables into integers to preserve their ordering.

Major	Year of Study	Study Hours /Week	Exam Score %	Stress Level
ENGR	1	12	65	Medium
COMP	1	3	61	N/A
COMP	3	15	N/A	High
AIML	2	20	81	Medium
ENGR	4	14	64	Low

Major_ ENGR	Major_ COMP	Major _AIML	Major_ MATH	Year of Study	Study Hours /Week	Exam Score %	Stress Level
1	0	0	0	1	12	65	**2**
0	1	0	0	1	3	61	**N/A**
0	1	0	0	3	15	N/A	**2**
0	0	1	0	2	20	81	**1**
1	0	0	0	4	14	64	**0**

Dimensionality reduction

For some datasets, we may wish to combine variables to reduce dimensionality, to visualize data better, or to remove redundancy.

Principal component analysis (PCA) finds new variables that are optimal linear combinations of the original variables, see examples here.

Major	Year of Study	Study Hours /Week	Exam Score %	Stress Level
ENGR	1	12	65	Medium
COMP	1	3	61	N/A
COMP	3	15	N/A	High
AIML	2	20	81	Medium
ENGR	4	14	64	Low

Today's Lecture

What is EDA? ✅

EDA methods by data type ✅

Step-by-step EDA ✅

Considerations in EDA

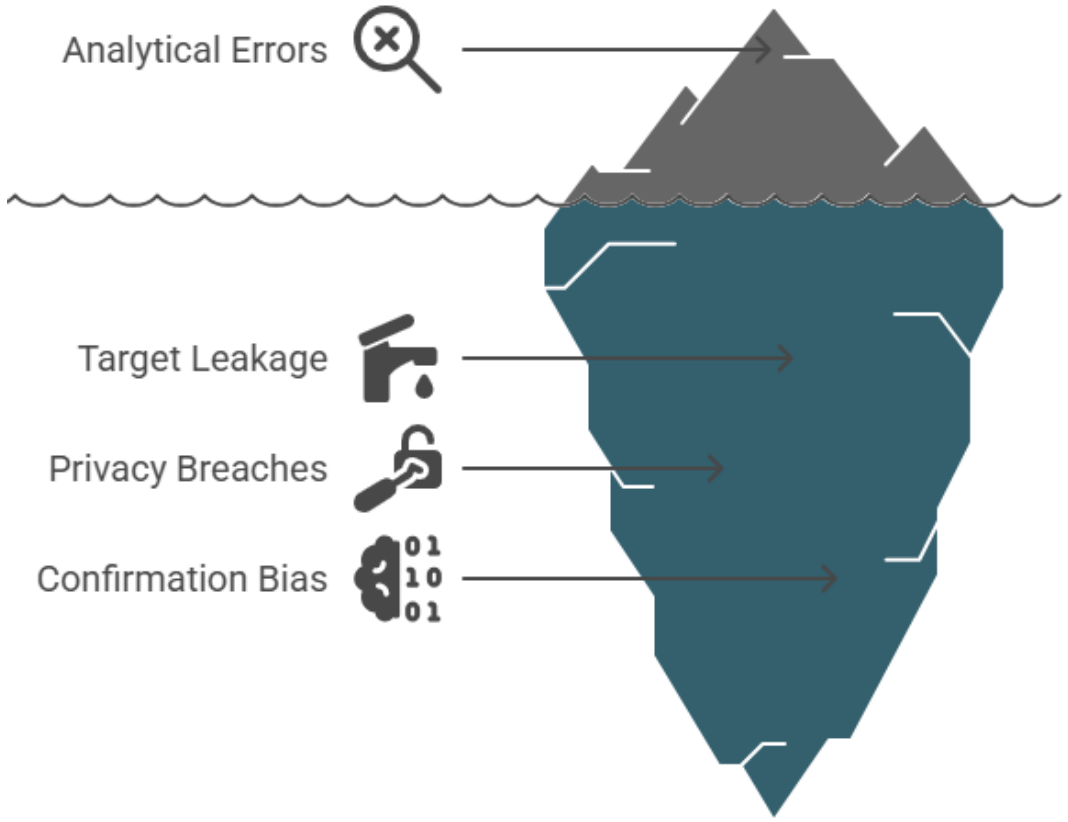
Target Leakage

The target variable (or other future variables) must not influence any data preprocessing steps.

<u>Common mistakes</u>

- Dimensionality reduction using the target variable
  - e.g. including the target when running PCA or feature selection
- Imputation using the target variable
  - e.g. predicting missing feature values using the target as a predictor
- Data transformations fitted using the entire dataset
  - e.g. scaling or preprocessing before the train/test split

Only use information that would be available when making a real prediction.



Privacy Breaches

Exploratory Data Analysis must not reveal personally identifiable information. Even when individual records are not shown, aggregated statistics can still expose individuals.

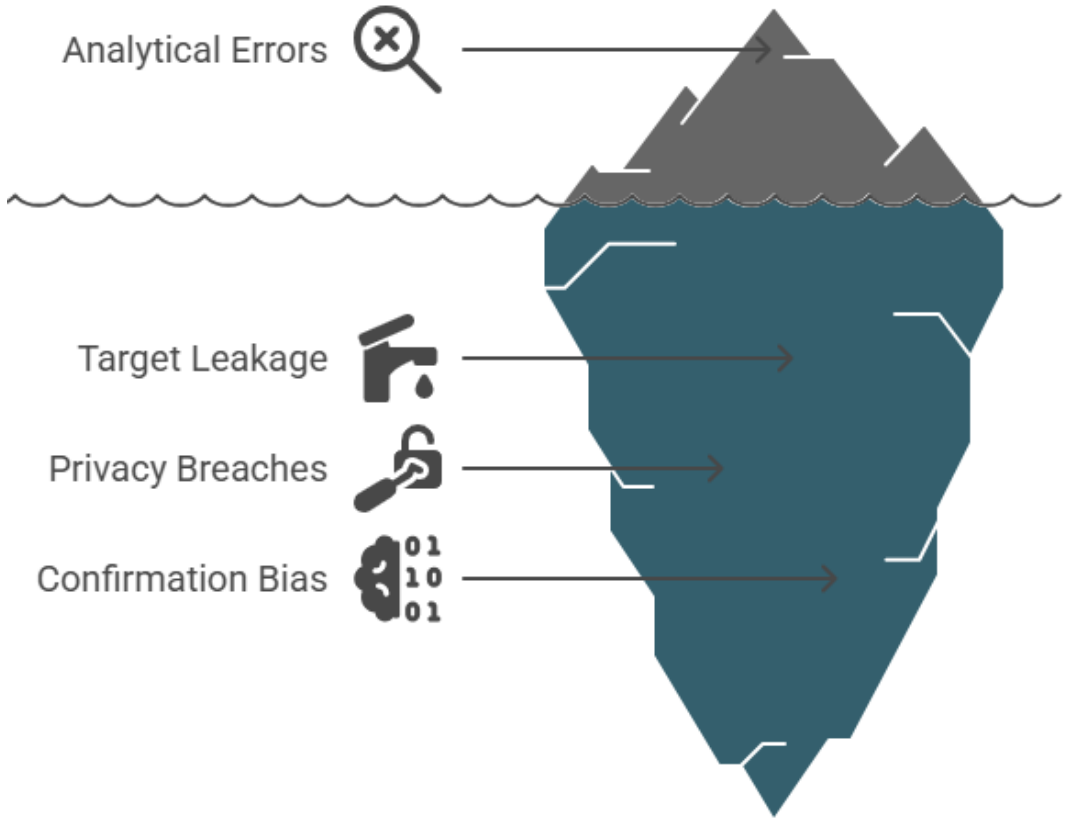
Example

- Suppose you create a boxplot of employee salaries by postcode.
- If only one employee lives in a particular postcode, the plot effectively reveals that person's salary.

When this happens most often

- Small datasets
- Highly specific groupings (postcode, age, department, etc.)

This is why EDA cannot always be fully automated and requires human judgement.



Confirmation Bias

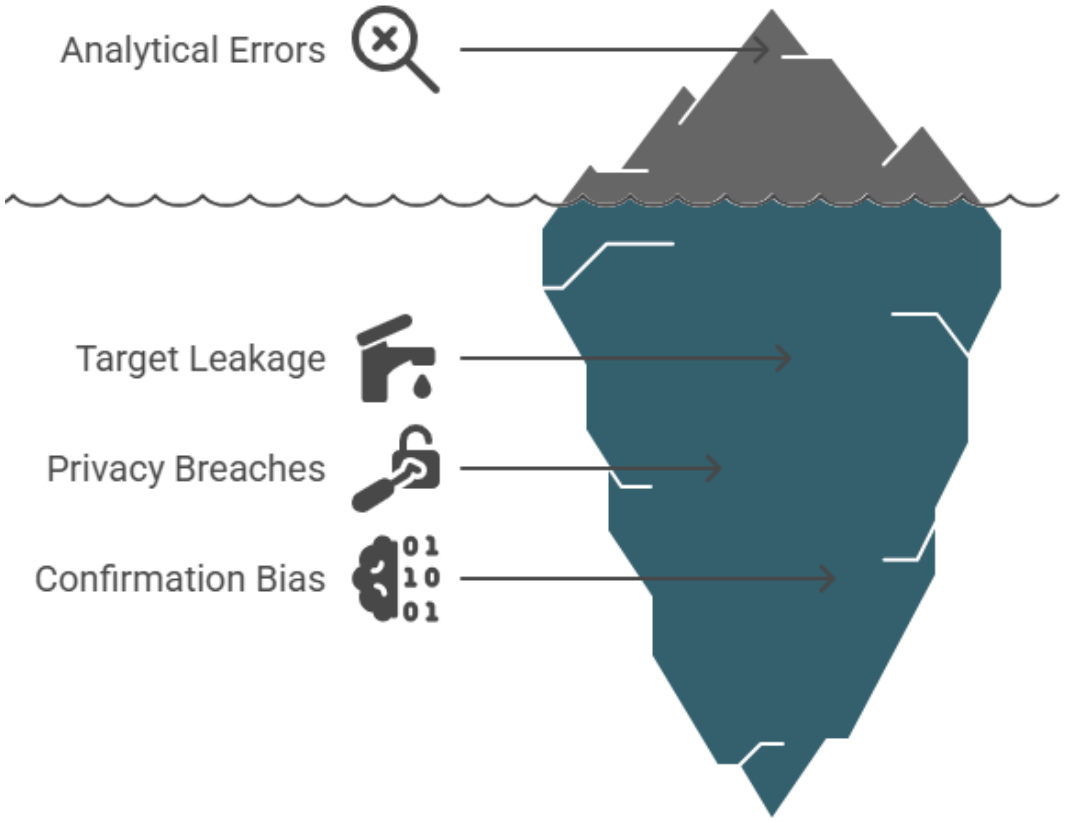
Do not search the data only for patterns that support your expectations or hypothesis. It is easy to notice patterns that confirm what you already believe, and ignore those that contradict it.

<u>Example</u>

Suppose you believe that students who study more achieve higher exam scores. During EDA you might:

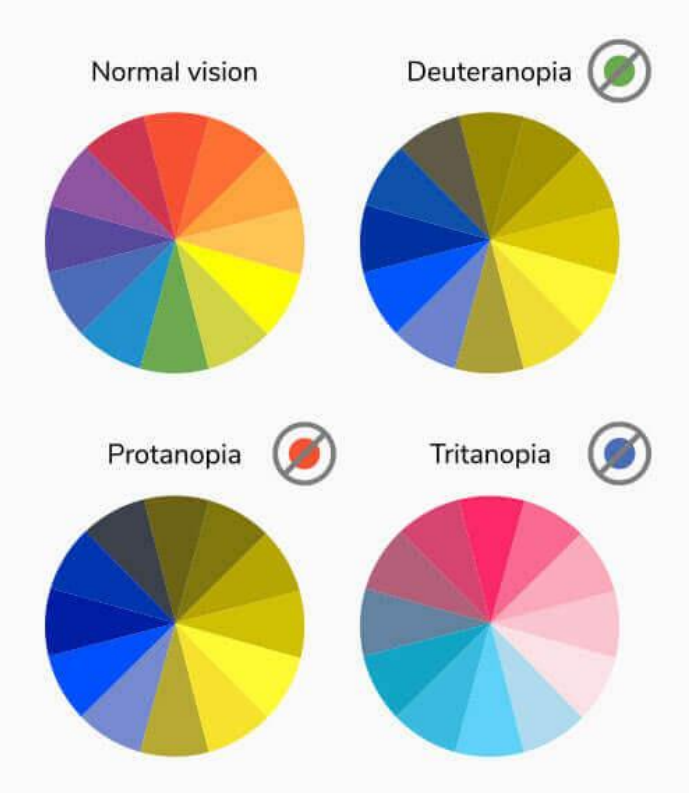
- Focus only on plots showing a positive relationship
- Ignore outliers
- Ignore other variables that might influence exam score

EDA should be applied without bias, and aim to **discover patterns, not confirm them.**



Accessibility

- Ensure your charts and plots are colour-blind friendly! Colour-blindness affects 1 in 12 men and 1 in 200 women[1].
- Red and green together ❌
- Consider using more than just colour – scatter plots can have different shapes, use text and labels
- Test out what your colour palette looks like to people with colour-blindness here



1. https://www.colourblindawareness.org/colour-blindness/

Today’s Lecture

- What is EDA? ✅
- EDA methods by data type ✅
- Step-by-step EDA ✅
- Considerations in EDA ✅